# Linear Graphs 1: Tables and the Equation

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**KEY WORDS** table of values *(tūtohi uara)* · gradient *(rōnaki)* · y-intercept *(pūtahi-y)* · y = mx + c

## A Reading a table ●○○

**Remember:** In a table for a straight line, $y$ changes by the same amount each time $x$ goes up by 1. That amount is the gradient $m$. The value of $y$ when $x = 0$ is the $y$-intercept $c$.

**1.** The table shows some of the points on a straight line.

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|
| $y$ | $-4$ | $-1$ | $2$ | $5$ | $8$ | $11$ |

**a)** What is $y$ when $x = 2$? _____

**b)** Each time $x$ goes up by 1, what happens to $y$? _____

**c)** What is $y$ when $x = 0$? _____

**d)** Continue the pattern: what is $y$ when $x = 4$? _____

**e)** Write the equation of the line. _____

**2.** The table shows some of the points on a straight line.

| $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|
| $y$ | $7$ | $5$ | $3$ | $1$ | $-1$ | $-3$ |

**a)** What is $y$ when $x = 3$? _____

**b)** Each time $x$ goes up by 1, what happens to $y$? _____

**c)** What is $y$ when $x = 0$? _____

**d)** Continue the pattern: what is $y$ when $x = 5$? _____

**e)** Write the equation of the line. _____

## B A table from the equation ●○○

**Remember:** Put each value of $x$ into the equation to find $y$. Take care with negatives: $2 \times (-3) = -6$.

**3.** Complete the table of values for each line.

**a)** $y = 2x - 3$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

**b)** $y = -x + 4$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

**c)** $y = 4 - 3x$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

**d)** $y = \frac{1}{2}x + 1$

| $x$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $y$ | | | | | |



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**e)** $y = 3x$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

**f)** $y = -\frac{1}{2}x + 3$

| $x$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

## C The equation from a table ●●○

**Remember:** The gradient is how much $y$ changes for each step of 1 in $x$. Read $c$ from the column where $x = 0$. Then write $y = mx + c$.

**4.** Each table shows points on a straight line. Write the equation of the line.

**a)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $-3$ | $-1$ | $1$ | $3$ | $5$ |

Equation: _____

**b)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $-10$ | $-7$ | $-4$ | $-1$ | $2$ |

Equation: _____

**c)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $5$ | $4$ | $3$ | $2$ | $1$ |

Equation: _____

**d)**

| $x$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $y$ | $5$ | $3$ | $1$ | $-1$ | $-3$ |

Equation: _____

**e)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $-8$ | $-4$ | $0$ | $4$ | $8$ |

Equation: _____

**f)**

| $x$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $y$ | $-6$ | $-5$ | $-4$ | $-3$ | $-2$ |

Equation: _____

## D When the table does not show x = 0 ●●○

**Remember:** Find $m$ first: the change in $y$ divided by the change in $x$. Then put one point of the table into $y = mx + c$ and solve for $c$.

**5.** Write the equation of each line.

**a)**

| $x$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|
| $y$ | $7$ | $9$ | $11$ | $13$ |

Equation:

**b)**

| $x$ | $3$ | $4$ | $5$ | $6$ |
|---|---|---|---|---|
| $y$ | $8$ | $11$ | $14$ | $17$ |

Equation:

**c)**

| $x$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|
| $y$ | $7$ | $5$ | $3$ | $1$ |

Equation:

**d)**

| $x$ | $1$ | $3$ | $5$ | $7$ |
|---|---|---|---|---|
| $y$ | $1$ | $5$ | $9$ | $13$ |

Equation:

**e)**

| $x$ | $2$ | $4$ | $6$ | $8$ |
|---|---|---|---|---|
| $y$ | $4$ | $2$ | $0$ | $-2$ |

Equation:

**f)**

| $x$ | $2$ | $4$ | $6$ | $8$ |
|---|---|---|---|---|
| $y$ | $8$ | $14$ | $20$ | $26$ |

Equation:


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## E Fractions, and tables that are not lines ●●●

> **Remember:** A gradient can be a fraction: if $y$ goes up by 3 while $x$ goes up by 2, the gradient is $\frac{3}{2}$. A table shows a straight line only when the $y$ values change by the same amount every step.

**6.** Write the equation of each line. Give the gradient as a fraction.

**a)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $0$ | $0.5$ | $1$ | $1.5$ | $2$ |

Equation:

**b)**

| $x$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|
| $y$ | $-3$ | $0$ | $3$ | $6$ |

Equation:

**c)**

| $x$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $y$ | $5$ | $4$ | $3$ | $2$ | $1$ |

Equation:

**d)**

| $x$ | $-3$ | $0$ | $3$ | $6$ |
|---|---|---|---|---|
| $y$ | $-1$ | $1$ | $3$ | $5$ |

Equation:

**7.** Does each table show a straight line? Answer yes or no, with a reason.

**a)** Table A

| $x$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|
| $y$ | $1$ | $3$ | $5$ | $7$ |

**b)** Table B

| $x$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|
| $y$ | $1$ | $4$ | $9$ | $16$ |

**c)** Table C

| $x$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|
| $y$ | $5$ | $4$ | $3$ | $2$ |

## F Tables in real situations ●●●

> **Remember:** The letters change but the idea is the same: the gradient is the change for each step, and the intercept is the starting value.

**8.** A taxi company charges a fixed amount plus a rate per kilometre. The table shows the fare $F$ in dollars for a trip of $d$ kilometres.

| $d$ (km) | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $F$ ($) | $5$ | $8$ | $11$ | $14$ | $17$ |

**a)** Write the equation that connects $F$ and $d$.

**b)** Find $F$ when $d = 7$.

**c)** Find $d$ when $F = 35$.

**d)** What does the number 5 tell you in this situation?

**e)** What does the number 3 tell you in this situation?


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**9.** A candle is lit. The table shows its height $h$ in centimetres after $t$ hours.

| $t$ (hours) | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $h$ (cm) | 20 | 18 | 16 | 14 | 12 |

**a)** Write the equation that connects $h$ and $t$.

**b)** Find $h$ when $t = 7$.

**c)** Find $t$ when $h = 0$.

**d)** What does the number 20 tell you in this situation?

**e)** What does the number 2 tell you in this situation?

## Extension

**10.** This table shows points on a straight line. Find its equation.

| $x$ | 4 | 7 | 10 | 13 |
|---|---|---|---|---|
| $y$ | 5 | 11 | 17 | 23 |

**11.** Two straight lines are shown by these tables. For which value of $x$ do the two lines have the same $y$ value? Find the equations first.

**a)** Line A

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 1 | 3 | 5 | 7 |

**b)** Line B

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 7 | 6 | 5 | 4 |

**c)** Same $y$ when

**12.** The equation of a line is $y = 4x - 3$. Find the value of $x$ that gives $y = 101$.


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# Linear Graphs 1: Tables and the Equation

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**KEY WORDS** table of values *(tūtohi uara)* · gradient *(rōnaki)* · y-intercept *(pūtahi-y)* · y = mx + c

## A Reading a table ●○○

**Remember:** In a table for a straight line, $y$ changes by the same amount each time $x$ goes up by 1. That amount is the gradient $m$. The value of $y$ when $x = 0$ is the $y$-intercept $c$.

**1.** The table shows some of the points on a straight line.

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|---|
| $y$ | $-9$ | $-5$ | $-1$ | $3$ | $7$ | $11$ |

**a)** What is $y$ when $x = 2$? _____

**b)** Each time $x$ goes up by 1, what happens to $y$? _____

**c)** What is $y$ when $x = 0$? _____

**d)** Continue the pattern: what is $y$ when $x = 4$? _____

**e)** Write the equation of the line. _____

**2.** The table shows some of the points on a straight line.

| $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|---|
| $y$ | $11$ | $8$ | $5$ | $2$ | $-1$ | $-4$ |

**a)** What is $y$ when $x = 2$? _____

**b)** Each time $x$ goes up by 1, what happens to $y$? _____

**c)** What is $y$ when $x = 0$? _____

**d)** Continue the pattern: what is $y$ when $x = 5$? _____

**e)** Write the equation of the line. _____

## B A table from the equation ●○○

**Remember:** Put each value of $x$ into the equation to find $y$. Take care with negatives: $2 \times (-3) = -6$.

**3.** Complete the table of values for each line.

**a)** $y = 3x - 2$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

**b)** $y = -x + 5$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

**c)** $y = 6 - 2x$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

**d)** $y = \frac{1}{2}x - 1$

| $x$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $y$ | | | | | |



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**e)** $y = -2x$

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

**f)** $y = -\frac{1}{2}x + 2$

| $x$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $y$ | | | | | |

## C The equation from a table ●●○

**Remember:** The gradient is how much $y$ changes for each step of 1 in $x$. Read $c$ from the column where $x = 0$. Then write $y = mx + c$.

**4.** Each table shows points on a straight line. Write the equation of the line.

**a)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $-5$ | $-2$ | $1$ | $4$ | $7$ |

Equation: _____

**b)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $-9$ | $-7$ | $-5$ | $-3$ | $-1$ |

Equation: _____

**c)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $6$ | $5$ | $4$ | $3$ | $2$ |

Equation: _____

**d)**

| $x$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $y$ | $7$ | $4$ | $1$ | $-2$ | $-5$ |

Equation: _____

**e)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $-10$ | $-5$ | $0$ | $5$ | $10$ |

Equation: _____

**f)**

| $x$ | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $y$ | $-8$ | $-7$ | $-6$ | $-5$ | $-4$ |

Equation: _____

## D When the table does not show x = 0 ●●○

**Remember:** Find $m$ first: the change in $y$ divided by the change in $x$. Then put one point of the table into $y = mx + c$ and solve for $c$.

**5.** Write the equation of each line.

**a)**

| $x$ | $2$ | $3$ | $4$ | $5$ |
|---|---|---|---|---|
| $y$ | $8$ | $11$ | $14$ | $17$ |

Equation:

**b)**

| $x$ | $3$ | $4$ | $5$ | $6$ |
|---|---|---|---|---|
| $y$ | $3$ | $5$ | $7$ | $9$ |

Equation:

**c)**

| $x$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|
| $y$ | $7$ | $4$ | $1$ | $-2$ |

Equation:

**d)**

| $x$ | $1$ | $3$ | $5$ | $7$ |
|---|---|---|---|---|
| $y$ | $1$ | $7$ | $13$ | $19$ |

Equation:

**e)**

| $x$ | $2$ | $4$ | $6$ | $8$ |
|---|---|---|---|---|
| $y$ | $7$ | $3$ | $-1$ | $-5$ |

Equation:

**f)**

| $x$ | $2$ | $4$ | $6$ | $8$ |
|---|---|---|---|---|
| $y$ | $5$ | $9$ | $13$ | $17$ |

Equation:


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## E Fractions, and tables that are not lines ●●●

> **Remember:** A gradient can be a fraction: if $y$ goes up by 3 while $x$ goes up by 2, the gradient is $\frac{3}{2}$. A table shows a straight line only when the $y$ values change by the same amount every step.

**6.** Write the equation of each line. Give the gradient as a fraction.

**a)**

| $x$ | $-2$ | $-1$ | $0$ | $1$ | $2$ |
|---|---|---|---|---|---|
| $y$ | $-2$ | $-1.5$ | $-1$ | $-0.5$ | $0$ |

Equation:

**b)**

| $x$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|
| $y$ | $-2$ | $1$ | $4$ | $7$ |

Equation:

**c)**

| $x$ | $-4$ | $-2$ | $0$ | $2$ | $4$ |
|---|---|---|---|---|---|
| $y$ | $4$ | $3$ | $2$ | $1$ | $0$ |

Equation:

**d)**

| $x$ | $-3$ | $0$ | $3$ | $6$ |
|---|---|---|---|---|
| $y$ | $-3$ | $-1$ | $1$ | $3$ |

Equation:

**7.** Does each table show a straight line? Answer yes or no, with a reason.

**a)** Table A

| $x$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|
| $y$ | $1$ | $2$ | $4$ | $8$ |

**b)** Table B

| $x$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|
| $y$ | $3$ | $5$ | $7$ | $9$ |

**c)** Table C

| $x$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|
| $y$ | $10$ | $7$ | $4$ | $1$ |

## F Tables in real situations ●●●

> **Remember:** The letters change but the idea is the same: the gradient is the change for each step, and the intercept is the starting value.

**8.** A plumber charges a call-out fee plus an hourly rate. The table shows the cost $C$ in dollars of a job that takes $h$ hours.

| $h$ **(hours)** | $0$ | $1$ | $2$ | $3$ | $4$ |
|---|---|---|---|---|---|
| $C$ **(\$)** | $60$ | $100$ | $140$ | $180$ | $220$ |

**a)** Write the equation that connects $C$ and $h$.

**b)** Find $C$ when $h = 6$.

**c)** Find $h$ when $C = 380$.

**d)** What does the number 60 tell you in this situation?

**e)** What does the number 40 tell you in this situation?


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**9.** A tank is being drained. The table shows the volume of water $V$ in litres after $t$ minutes.

| $t$ (minutes) | 0 | 1 | 2 | 3 | 4 |
|---|---|---|---|---|---|
| $V$ (litres) | 500 | 475 | 450 | 425 | 400 |

**a)** Write the equation that connects $V$ and $t$.

**b)** Find $V$ when $t = 12$.

**c)** Find $t$ when $V = 0$.

**d)** What does the number 500 tell you in this situation?

**e)** What does the number 25 tell you in this situation?

## Extension

**10.** This table shows points on a straight line. Find its equation.

| $x$ | 5 | 8 | 11 | 14 |
|---|---|---|---|---|
| $y$ | 11 | 20 | 29 | 38 |

**11.** Two straight lines are shown by these tables. For which value of $x$ do the two lines have the same $y$ value? Find the equations first.

**a)** Line A

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | $-2$ | 1 | 4 | 7 |

**b)** Line B

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 10 | 9 | 8 | 7 |

**c)** Same $y$ when

**12.** The equation of a line is $y = 5x + 2$. Find the value of $x$ that gives $y = 102$.


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